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Started on Wednesday, 20 November 2024, 9:58 AM

State Finished

Completed on Wednesday, 20 November 2024, 10:19 AM

Time taken 20 mins 36 secs

Grade 17.50 out of 21.00 (83%)

Question **1**

Correct

Mark 2.00 out of 2.00

Match up the following statements about hashing

What does closed address hashing use to resolve collisions?

linked lists



What does open address hashing use to resolve collisions?

rehashing



Your answer is correct.

The correct answer is:

What does closed address hashing use to resolve collisions? → linked lists,

What does open address hashing use to resolve collisions? → rehashing

Question **2**

Correct

Mark 4.00 out of 4.00

Imagine we are trying to insert the elements 1, 5, 7, 2, 3, 0, 4, 10 (in that order) into a hash table of size 10 using an open address hashing scheme with linear probing. Our hash function is given by $h(x) = (x(x + 1) + x^2) \bmod 10$. Given this, specify the location of each key in the hashtable once they have all been inserted.

Key 0 is stored at location ✓Key 1 is stored at location ✓Key 2 is stored at location ✓Key 3 is stored at location ✓Key 4 is stored at location ✓Key 5 is stored at location ✓Key 7 is stored at location ✓Key 10 is stored at location ✓

Your answer is correct.

The correct answer is:

Imagine we are trying to insert the elements 1, 5, 7, 2, 3, 0, 4, 10 (in that order) into a hash table of size 10 using an open address hashing scheme with linear probing. Our hash function is given by $h(x) = (x(x + 1) + x^2) \bmod 10$. Given this, specify the location of each key in the hashtable once they have all been inserted.

Key 0 is stored at location [2]

Key 1 is stored at location [3]

Key 2 is stored at location [0]

Key 3 is stored at location [1]

Key 4 is stored at location [7]

Key 5 is stored at location [5]

Key 7 is stored at location [6]

Key 10 is stored at location [4]

Question **3**

Correct

Mark 1.00 out of 1.00

In closed address hashing we will always find the key by going to the position in the hash table that is given by the hash function.

Select one:

- ☒ True ✓
- ☐ False

If we go to that position in the has table we will find the key although it will actually be in a linked list accessed from that position.

The correct answer is 'True'.

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following are desirable properties of hash functions?

- ☒ a. Be easily understandable ✗
- ☒ b. Good spread over the available indices ✓
- ☒ c. Simple to compute ✗
- ☒ d. Result in few collisions ✓

Your answer is correct.

The correct answers are:

Good spread over the available indices,

Result in few collisions

Question **5**

Incorrect

Mark 0.00 out of 1.00

Phephile has written her own implementation of hashing with separate chaining. However, she decides to modify the chaining scheme by keeping each linked list in sorted order. Given this change, and denoting the load factor as α , what is the average time complexity of her data structure?

- ☐ a. Insertion: $O(1)$; Search: $O(1 + \lg \alpha)$; Deletion $O(1 + \lg \alpha)$
- ☒ b. Insertion: $O(1 + \lg \alpha)$; Search: $O(1 + \lg \alpha)$; Deletion $O(1 + \lg \alpha)$ ✖
- ☐ c. Insertion: $O(1 + \alpha)$; Search: $O(1 + \alpha)$; Deletion $O(1 + \alpha)$
- ☐ d. Insertion: $O(1)$; Search: $O(1 + \alpha)$; Deletion $O(1 + \alpha)$

Your answer is incorrect.

The correct answer is:

Insertion: $O(1 + \alpha)$; Search: $O(1 + \alpha)$; Deletion $O(1 + \alpha)$

Question 6

Not answered

Marked out of 1.00

Suppose we are storing pet's names (and associated information) in a hash table using open address hashing.

After entering some information we get the hash table shown below.

Suppose now that we need to delete Flopsy from the table.

Which of the tags below should be used to fill the space left by removing Flopsy?

Drag the correct rectangle into position in the hash table.

0	"jeff"
1	nil
2	"mutt"
3	"flopsy"
4	"pina"
5	"becca"
6	"fred"
7	"spot"
8	nil

Nil

Obsolete

Deleted

Your answer is incorrect.

Question 7

Correct

Mark 1.00 out of 1.00

Suppose we are storing pet's names (and associated information) in a hash table using open address hashing.

After entering some information we get the hash table shown below.

Suppose now that we need to insert Rover into the table. Rover's hash code is 0.

Indicate where in the table, Rover will be stored.

(Note that the circle on the marker must fall into the slot you are trying to address in order to get selected into that slot).

0	"jeff"
1	nil
2	"mutt"
3	"flopsy"
4	"pina"
5	"becca"
6	"fred"
7	"spot"
8	nil



Your answer is correct.

Question 8

Correct

Mark 1.00 out of 1.00

In a valid Red Black Tree, a Black node can have Black children.

Select one:

☒ True ✓

☐ False

By the definition

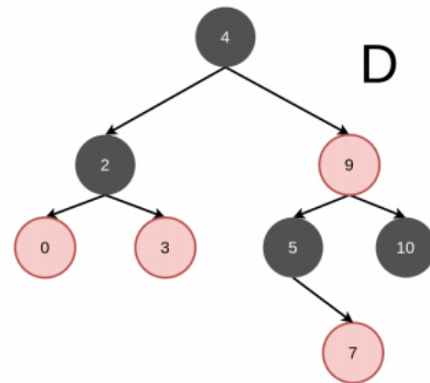
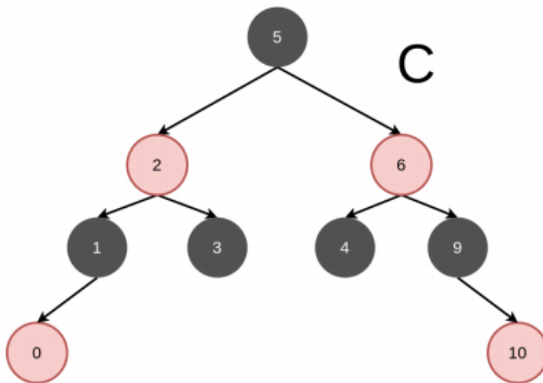
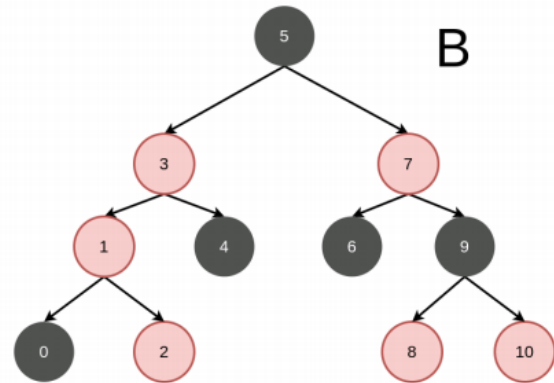
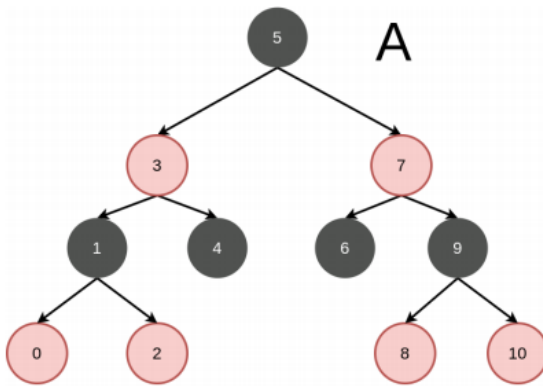
The correct answer is 'True'.

Question 9

Partially correct

Mark 0.50 out of 1.00

Which of the following below are valid red-black trees? (Null leaf nodes are not shown)



☒ A ✓

☐ B

☒ C ✗

☒ D ✓

☐ None of the above are valid red-black trees

Your answer is partially correct.

You have selected too many options.

The correct answers are:

A,

D

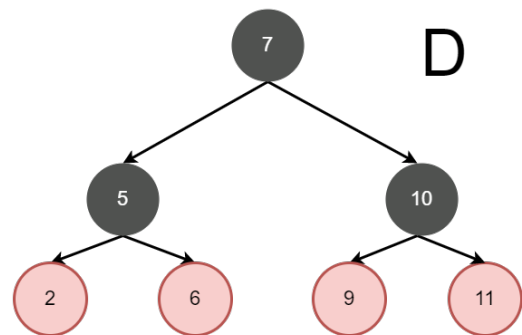
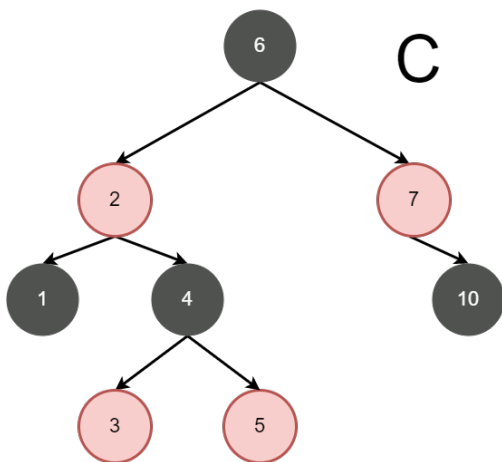
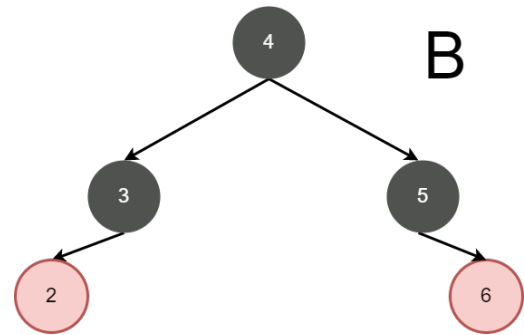
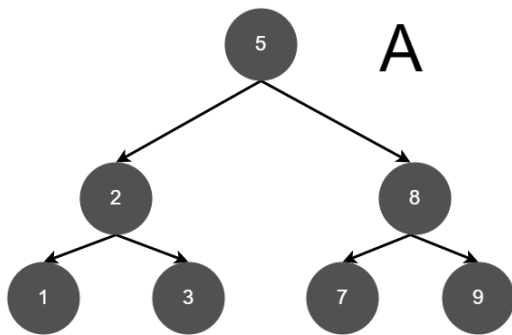
Question 10

Correct

Mark 1.00 out of 1.00

Which of the trees below are **invalid** red-black trees?

Note that for clarity the **nil** nodes are not shown but they should be assumed to be there.



- ☐ a. A
- ☐ b. B
- ☒ c. C ✓
- ☐ d. D

Your answer is correct.

The correct answer is:

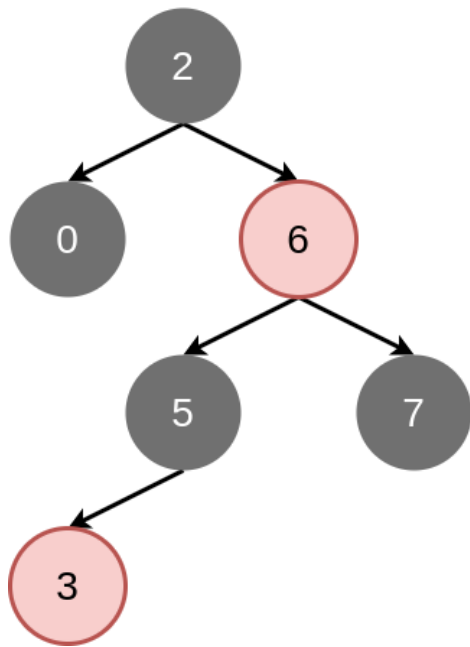
C

Question **11**

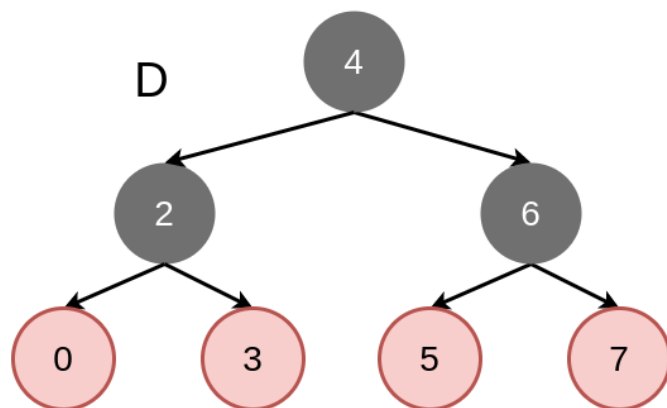
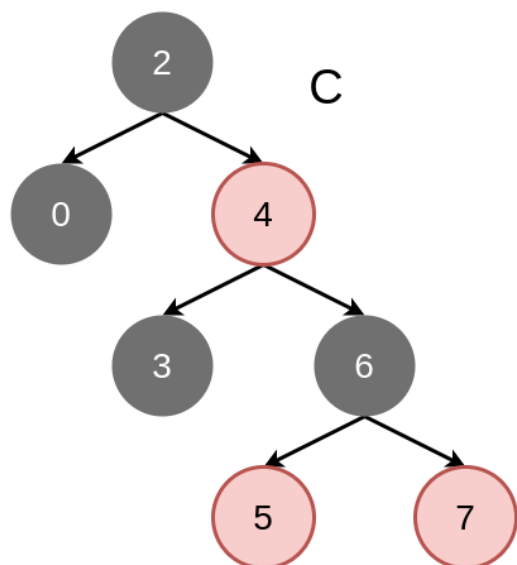
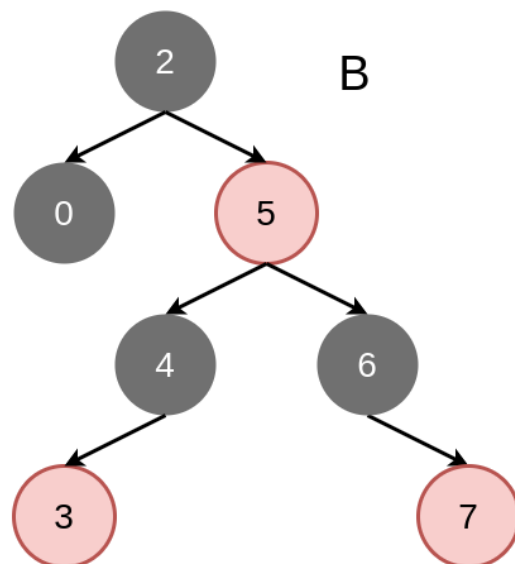
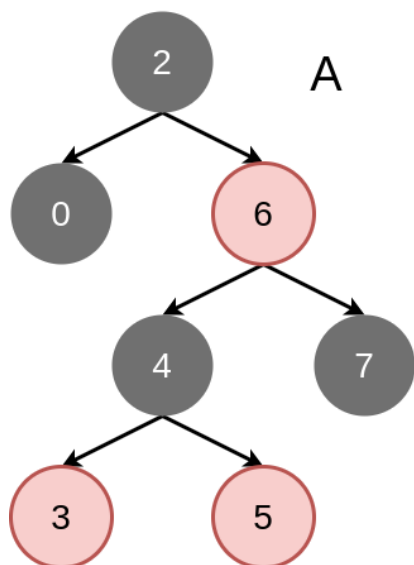
Correct

Mark 1.00 out of 1.00

Consider the red-black tree below and assume we wish to insert the value 4 into it. (Null leaf nodes are not shown)



Select the red-black tree below that results when 4 is inserted (using the red-black tree insertion algorithm).



- ☒ A ✓
☐ B
☐ C
☐ D
☐ None of the above

Your answer is correct.

The correct answer is:

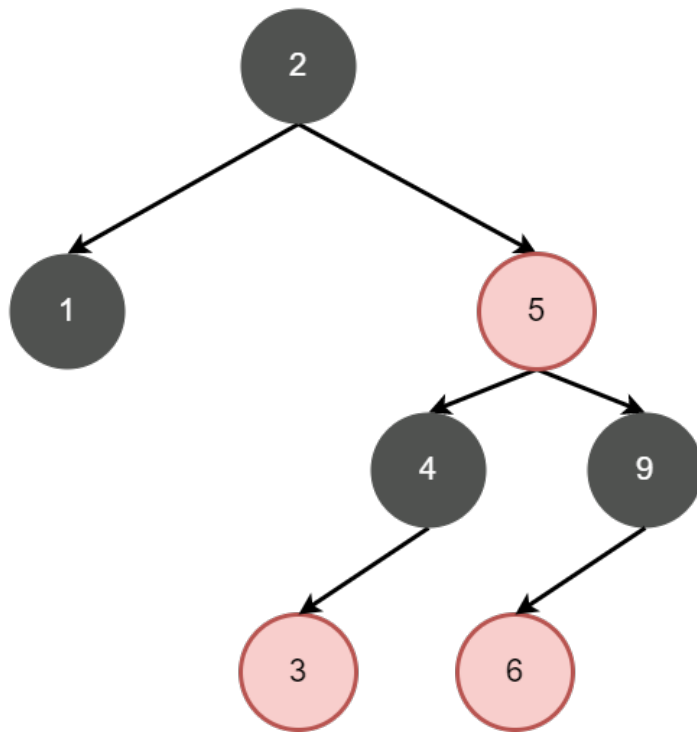
A

Question **12**

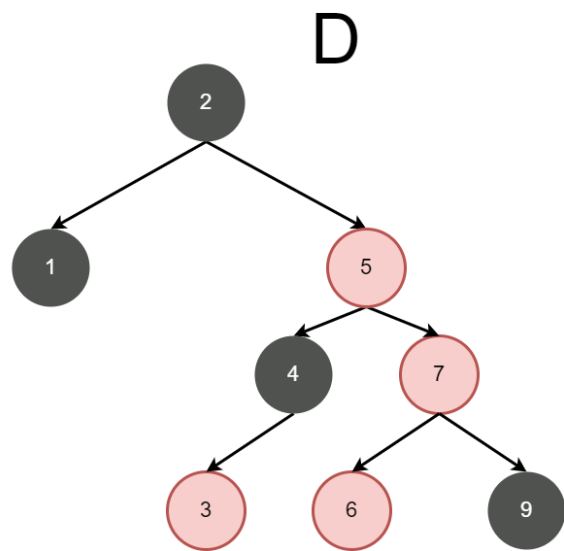
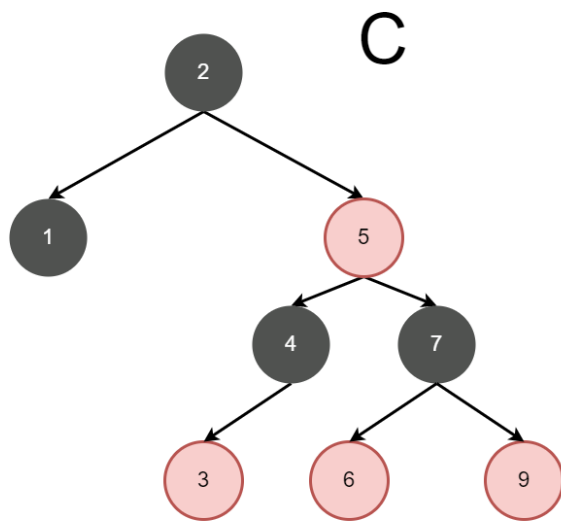
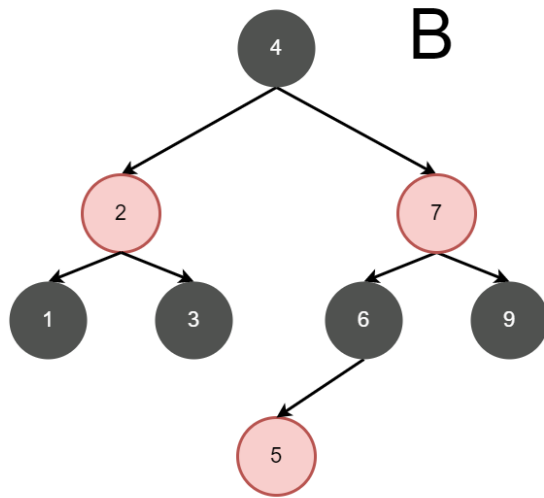
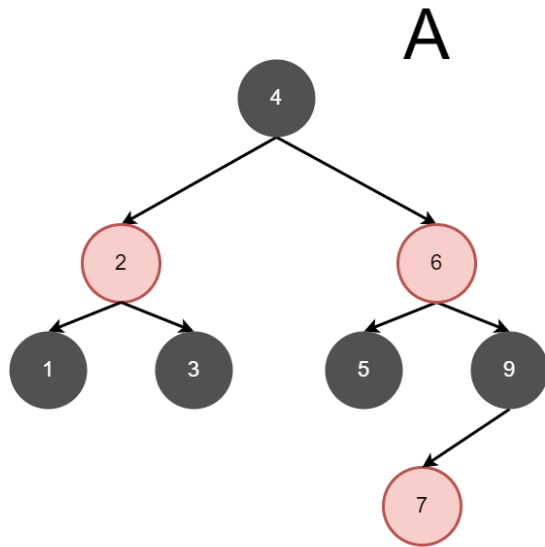
Correct

Mark 1.00 out of 1.00

Consider the red-black tree below and assume we wish to insert the value 7 into it.



Select the tree below that results when 7 is inserted.



- ☐ a. A
- ☐ b. B
- ☒ c. C ✓
- ☐ d. D

Your answer is correct.

The correct answer is:

C

Question **13**

Correct

Mark 2.00 out of 2.00

Which of the items below are features of Red Black Trees?

- ☒ a. They allow insertion in logarithmic time. ✓
- ☐ b. They can store fewer keys than a hash table
- ☐ c. They require quadratic time for deletion of a key value
- ☒ d. They allow very fast access to successor and/or predecessor values ✓
- ☐ e. They allow constant time searching for a given key
- ☒ f. They maintain the key data in order ✓ from notes

Your answer is correct.

The correct answers are:

They maintain the key data in order,

They allow very fast access to successor and/or predecessor values,

They allow insertion in logarithmic time.

Question **14**

Partially correct

Mark 2.00 out of 3.00

Match the statements below about Hash Tables and Red Black Trees.

A hash table is an example of a	dictionary	✓
Direct Addressing is not efficient if	store many keys that have the same value.	✗
A red black tree can only be used to	store key values that have some inherent ordering	✓

Your answer is partially correct.

You have correctly selected 2.

The correct answer is:

A hash table is an example of a → dictionary, Direct Addressing is not efficient if → the key space is very large.,

A red black tree can only be used to → store key values that have some inherent ordering

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